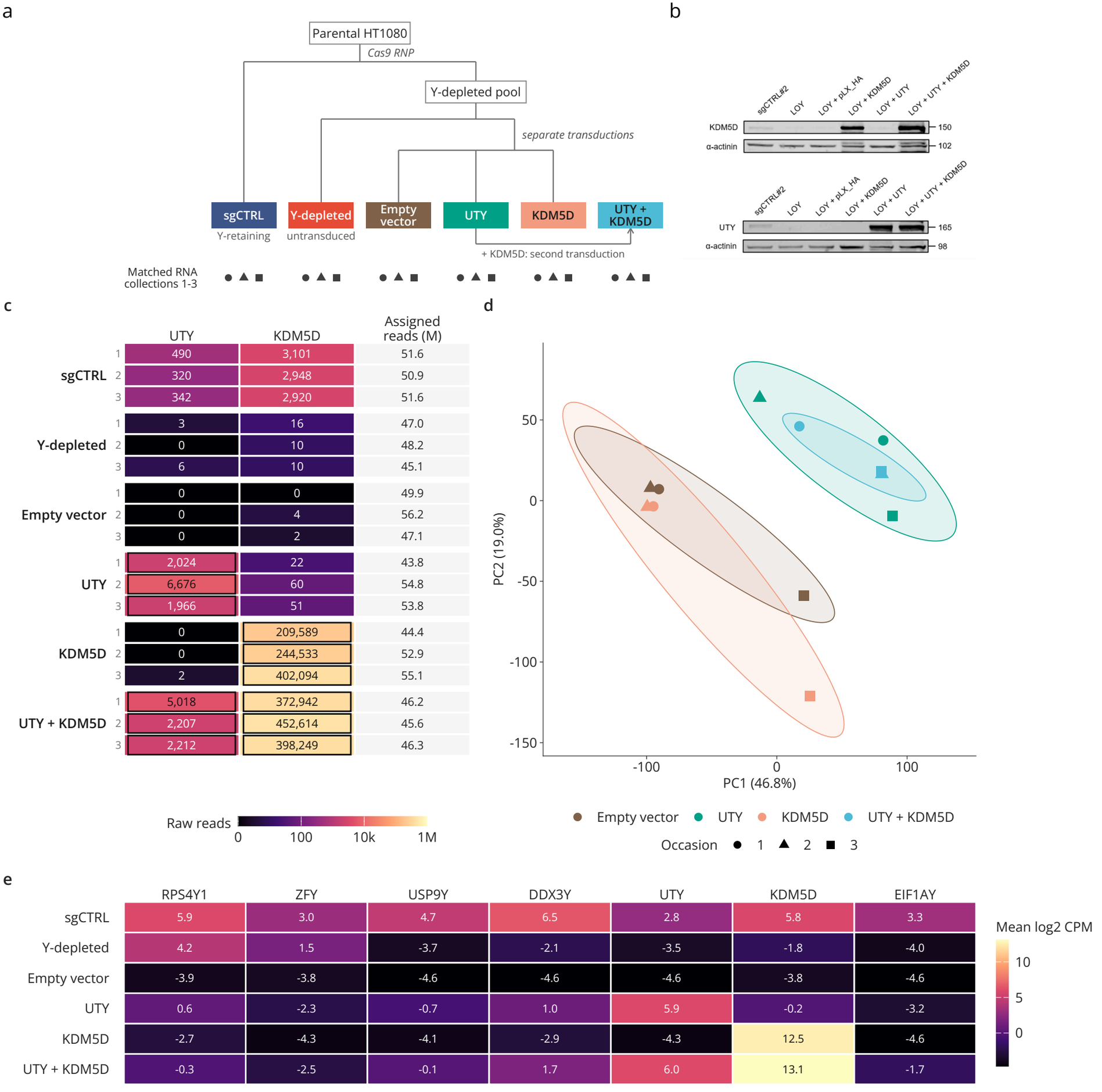

a
Parental HT1080
Cas9 RNP
Y-depleted pool
separate transductions
sgCTRL
Y-retaining
Y-depleted
untransduced
Empty vector
UTY
KDM5D
UTY + KDM5D
+ KDM5D: second transduction
Matched RNA collections 1-3
b
sgCTRL#2
LOY
LOY + pLX_HA
LOY + KDM5D
LOY + UTY
LOY + UTY + KDM5D
KDM5D
α-actinin
150
102
UTY
165
98
c
UTY
KDM5D
Assigned reads (M)
sgCTRL 1 490 3,101 51.6
sgCTRL 2 320 2,948 50.9
sgCTRL 3 342 2,920 51.6
Y-depleted 1 3 16 47.0
Y-depleted 2 0 10 48.2
Y-depleted 3 6 10 45.1
Empty vector 1 0 0 49.9
Empty vector 2 0 4 56.2
Empty vector 3 0 2 47.1
UTY 1 2,024 22 43.8
UTY 2 6,676 60 54.8
UTY 3 1,966 51 53.8
KDM5D 1 0 209,589 44.4
KDM5D 2 0 244,533 52.9
KDM5D 3 2 402,094 55.1
UTY + KDM5D 1 5,018 372,942 46.2
UTY + KDM5D 2 2,207 452,614 45.6
UTY + KDM5D 3 2,212 398,249 46.3
Raw reads
0
100
10k
1M
d
PC2 (19.0%)
PC1 (46.8%)
50
0
-50
-100
-150
-100
0
100
Empty vector
UTY
KDM5D
UTY + KDM5D
Occasion 1 2 3
e
RPS4Y1 ZFY USP9Y DDX3Y UTY KDM5D EIF1AY
sgCTRL 5.9 3.0 4.7 6.5 2.8 5.8 3.3
Y-depleted 4.2 1.5 -3.7 -2.1 -3.5 -1.8 -4.0
Empty vector -3.9 -3.8 -4.6 -4.6 -4.6 -3.8 -4.6
UTY 0.6 -2.3 -0.7 1.0 5.9 -0.2 -3.2
KDM5D -2.7 -4.3 -4.1 -2.9 -4.3 12.5 -4.6
UTY + KDM5D -0.3 -2.5 -0.1 1.7 6.0 13.1 -1.7
Mean log2 CPM
10
5
0